

CMPG 325 - Computer Networks

Technical Report

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TABLE OF CONTENTS

A. Executive Summary & System Overview .....

B. Network Design And Topologies ..... …….

C. Ip Addressing And Vlsm Subnet Plan .....

D. Core Network Services Deployment .....

E. Technical Implementation And Security Policies .....

F. Testing And Verification Results .....

G. Operational Challenges And Future Expansion .....

H. Conclusion .....

I. Software Tool And Project Profile .....

TECHNICAL REPORT FOR LEANO GAMING LOUNGE PROJECT

A. EXECUTIVE SUMMARY & SYSTEM OVERVIEW

I. INTRODUCTION AND PROJECT OBJECTIVE

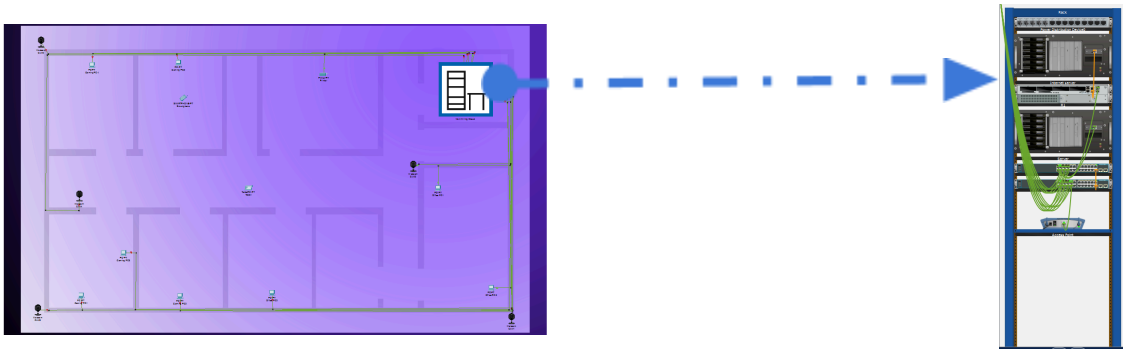
The Leano Gaming Lounge, situated in Mahikeng, required a robust, highly secure, and scalable local area network (LAN) infrastructure to support its modern entertainment operations. As digital entertainment venues increasingly rely on continuous uptime, secure customer Wi-Fi, and protected administrative systems, the underlying network must balance high-speed data delivery with strict security segmentation.

This project encompassed the complete engineering lifecycle, from initial requirements analysis and logical topology design to Cisco Packet Tracer simulation, deployment of core network services, and exhaustive functional verification. The primary objective was to engineer an enterprise-grade network within the assigned IPv4 base block (192.168.55.0/24) that seamlessly accommodates competitive gaming terminals, administrative offices, CCTV security monitoring, shared printing services, and public visitor Wi-Fi.

B. NETWORK DESIGN AND TOPOLOGIES

To meet the operational demands of the lounge while optimizing cost and performance, hierarchical network topologies were designed and simulated using Cisco Packet Tracer.

I. PHYSICAL TOPOLOGY

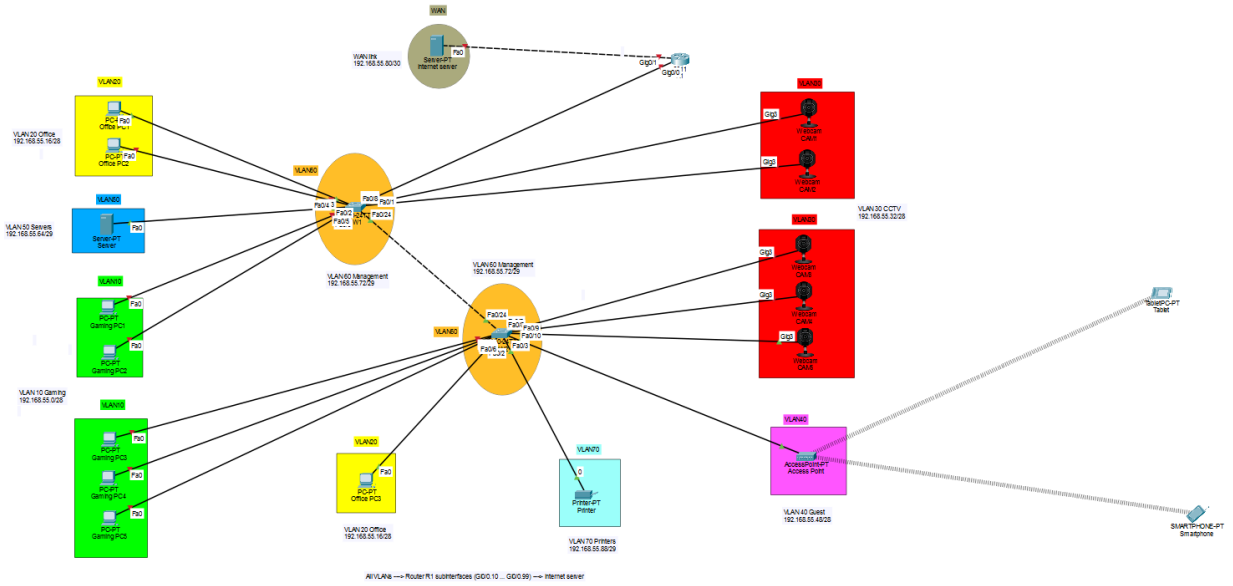


II. LOGICAL TOPOLOGY

Core Routing Layer: A Cisco 2911 router (R1) serves as the central default gateway and routing engine for the entire facility. Rather than provisioning individual physical router ports for each department, R1 utilizes a Router-on-a-Stick (ROAS) topology via 802.1Q subinterfaces. This approach maximizes routing efficiency and accommodates future VLAN additions without requiring hardware modifications.

Access Switching Layer: Two Cisco Catalyst 2960 switches (SW1 and SW2) are deployed to aggregate end devices geographically across the facility. SW1 anchors the server room, administrative offices, and primary gaming clusters, while SW2 supports secondary gaming stations, CCTV webcams, the network printer, and the guest access point.

Trunking Backbone: An 802.1Q trunk link established between port Fa0/24 on SW1 and port Fa0/24 on SW2 enables multi-switch VLAN propagation, ensuring that endpoints physically connected to different switches remain logically grouped within their respective departmental subnets.



C. IP ADDRESSING AND VLSM SUBNET PLAN

To prevent broadcast storms, isolate department traffic, and ensure efficient address allocation within the restricted /24 space (192.168.55.0/24), Variable Length Subnet Masking (VLSM) was implemented:

VLAN	Interface	Network	Mask	Usable range	Hosts needed	Capacity	Address assignment
10	Gaming	192.168.55.0	/28 (255.255.255.240)	.1 to .14	5	14	DHCP from server, pool .2 to .14
20	Office	192.168.55.16	/28 (255.255.255.240)	.17 to .30	4	14	DHCP from server, pool .18 to .30
30	CCTV	192.168.55.32	/28 (255.255.255.240)	.33 to .46	5	14	Static, .34 to .38
40	Guest	192.168.55.48	/28 (255.255.255.240)	.49 to .62	Up to 13	14	DHCP from router, pool .50 to .62
50	Servers	192.168.55.64	/29 (255.255.255.248)	.65 to .70	1	6	Static, server .66
60	Management	192.168.55.72	/29 (255.255.255.248)	.73 to .78	2	6	Static, SW1 .74, SW2 .75
70	Printer	192.168.55.88	/29 (255.255.255.248)	.89 to .94	1	6	Static, printer .90
99	Native	none	none	none	0	none	Unused
WAN	Router to Internet	192.168.55.80	/30 (255.255.255.252)	.81 to .82	2	2	Static, router .81, Internet server .82

D. CORE NETWORK SERVICES DEPLOYMENT

I. Internal Server (192.168.55.66):

- **DHCP Service:** Configured to dynamically lease IP addresses, default gateways, and DNS server parameters to endpoints across the Gaming (VLAN 10) and Office (VLAN 20) subnets.
- **DNS Service:** Authoritative internal name resolution mapping server.leano.local to 192.168.55.66.
- **HTTP Service:** Hosts internal intranet pages for staff management.

II. Internet Server (192.168.55.82):

- **HTTP Service:** Hosts the custom external web portal www.leanogaminglounge.com representing the lounge's public-facing digital presence.
- **DNS Service:** Simulates external public domain name resolution

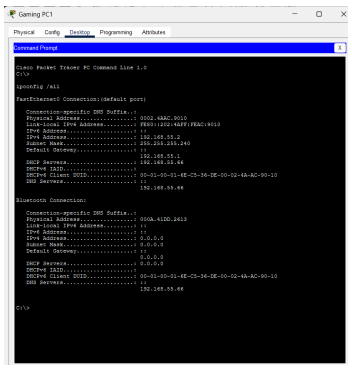
E. TECHNICAL IMPLEMENTATION AND SECURITY POLICIES

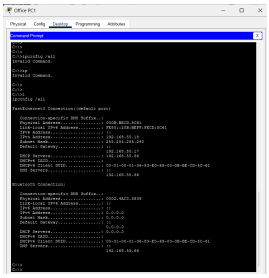
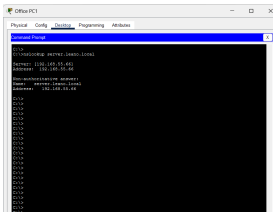
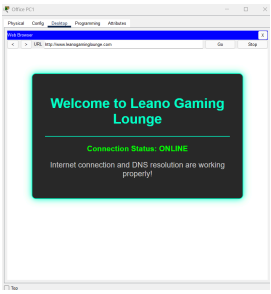
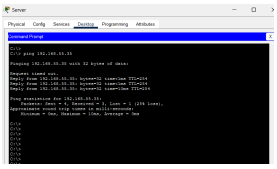
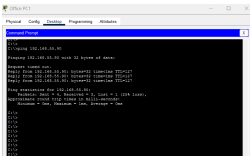
CCTV Traffic Segmentation: Cameras on VLAN 30 are governed by the CCTV-IN ACL applied inbound on subinterface Gig0/0.30. This policy permits camera traffic to reach its default gateway and the internal server but explicitly drops all packets destined for the administrative office network.

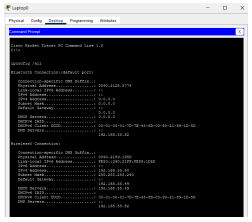
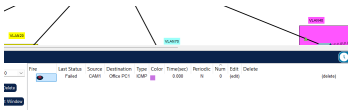
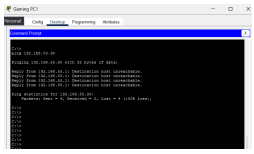
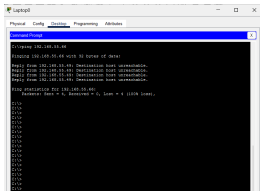
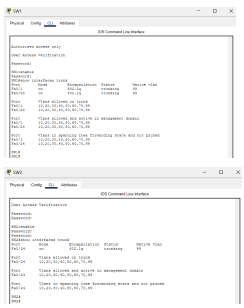
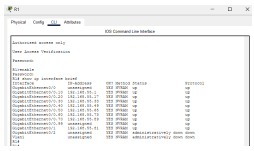
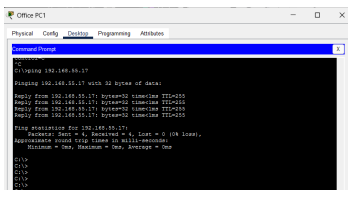
Guest Wi-Fi Isolation: An independent Wireless Access Point broadcasting SSID Leano-Guest is connected to SW2 (Fa0/10) on VLAN 40. Router R1 acts as the local DHCP server for guests. The GUEST-IN ACL ensures visitors can access the public internet and obtain local DHCP leases while being completely isolated from internal gaming, office, CCTV, and printer subnets.

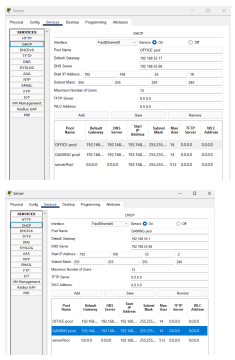
Administrative and Gaming Control: Dedicated ACLs (OFFICE-IN and GAMING-IN) prevent gaming terminals from accessing sensitive administrative printers or CCTV feeds while preserving high-speed internet and server access.

F. TESTING AND VERIFICATION RESULTS

Test ID	Category	Action / Test Description	Expected Result	Status	Evidence Reference
T01	DHCP Lease	Configure Gaming PC1 to DHCP.	Obtains IP in range 192.168.55.2–14 with DNS 192.168.55.66	PASS	

T02	DHCP Lease	Configure Office PC1 to DHCP.	Obtains IP in range 192.168.55.18–30 with gateway 192.168.55.17	PASS	
T03	Internal DNS	Execute nslookup server.leano.local on Office PC1.	Resolves successfully to 192.168.55.66	PASS	
T04	HTTP Web	Open www.leanogaminglounge.com from the Office PC browser.	Leano Gaming Lounge landing page renders successfully	PASS	
T05	Static IP	Ping CAM4 (192.168.55.35 on SW2) from Internal Server	Ping replies successfully across trunk link	PASS	
T06	Printer Reach	Ping Network Printer (192.168.55.90) from Office PC1.	Successful ping replies confirming cross-VLAN printing	PASS	

T06	Guest Wi-Fi	Connect Wireless Laptop to SSID Leano-Guest	Receives IP 192.168.55.50 from gateway 192.168.55.49	PASS	
T08	CCTV ACL	Ping Office PC1 (192.168.55.19) from CAM1 (192.168.55.34).	Blocked by CCTV-IN ACL (Destination Host Unreachable)	PASS	
T09	Gaming ACL	Ping Network Printer (192.168.55.90) from Gaming PC1.	Blocked by GAMING-IN ACL	PASS	
T10	Guest ACL	Ping Internal Server (192.168.55.66) from Guest Laptop.	Blocked by GUEST-IN ACL enforcing complete isolation	PASS	
T11	Trunking	Run show interfaces trunk on SW1 and SW2 CLI.	Inter-switch link (Fa0/24) status shows trunking under 802.1Q	PASS	
T12	ROAS Status	Run show ip interface brief on R1 CLI.	Subinterfaces Gig0/0.10 to .99 display up/up	PASS	
T13	Inter-VLAN	Ping Office PC3 (192.168.55.17) from Office PC1 (192.168.55.19).	Successful ICMP communication across switches	PASS	

T14	DHCP Binding	View active lease table on Internal Server GUI.	Logs active MAC-to-IP leases for Gaming and Office PCs	PASS	
T15	NVRAM Backup	Run copy running-config startup-config on R1, SW1, SW2.	Returns [OK]; configuration written to startup memory	PASS	

T01–T02 (DHCP Leases): Gaming and Office PCs automatically acquired correct IP leases from the Internal Server.

T03–T04 (DNS & HTTP): Internal DNS successfully resolved server.leano.local, and Office PCs successfully rendered the external web portal - www.leanogaminglounge.com.

T05–T07 (Static & Wireless Reachability): Pings from the Internal Server to CCTV webcams (CAM4) and from Office PCs to the Network Printer succeeded across trunk links. Wireless laptops successfully associated with Leano-Guest and obtained local leases from R1.

T08–T10 (Security Enforcement): Cross-segment security checks confirmed that CCTV webcams, gaming terminals, and guest Wi-Fi devices were successfully blocked from accessing unauthorized internal resources by router ACLs.

T11–T15 (Infrastructure Validation): Switch trunking (show interfaces trunk), router subinterface status (show ip interface brief), inter-VLAN routing, DHCP bindings, and NVRAM configuration backups (copy running-config startup-config) verified a stable, fully operational deployment

G. OPERATIONAL CHALLENGES AND FUTURE EXPANSION

I. WIRELESS MODULE INCOMPATIBILITY ON END DEVICES:

Issue: The mobile device added to simulate a guest visitor failed to detect or associate with Access Point Leano-Guest.
Resolution: Identified that default Packet Tracer laptop templates are provisioned with wired Ethernet cards (PT-LAPTOP-INT-1ETH). Powered down the virtual laptop, replaced the wired module with a WPC300N wireless PCMCIA card, repowered the device, and established a successful wireless association.

II. CROSS-SWITCH STATIC CCTV ROUTING FAILURES:

Issue: Initial pings from the Internal Server (SW1) to Webcam CAM4 (SW2) resulted in request timeouts.
Resolution: Discovered that while CAM4 had a valid static IP (192.168.55.37), its Default Gateway field was left blank (0.0.0.0). Populating the gateway field with 192.168.55.33 (R1 Gig0/0.30 gateway) immediately restored bidirectional traffic across the 802.1Q trunk.

III. DHCP POOL OVERLAPS:

Issue: Intermittent DHCP lease failures on gaming terminals during early staging.
Resolution: Re-audited VLSM boundaries to ensure DHCP scopes did not overlap with statically assigned gateway, server, or printer addresses.

IV. RECOMMENDATIONS FOR FUTURE EXPANSION

- **Link Redundancy & Spanning Tree (STP):** Introduce secondary trunk links between SW1 and SW2 and configure Rapid Spanning Tree Protocol (RSTP / IEEE 802.1w) or EtherChannel to eliminate single points of failure.
- **Dynamic Routing:** Transition from static routing and direct subinterface attachment to OSPF (Open Shortest Path First) as additional branch locations or multi-floor switches are integrated.
- **Port Security:** Implement Cisco Port Security on access layer switches (SW1 and SW2) to restrict port access to authorized MAC addresses, preventing unauthorized devices from connecting to administrative ports.
- **Encrypted Management:** Enforce SSHv2 for all switch and router CLI management sessions, replacing unencrypted Telnet access.

H. CONCLUSION

The Leano Gaming Lounge network implementation has been fully completed, tested, and verified. All operational requirements, security constraints, and technical challenges including 802.1Q trunking and guest Wi-Fi isolation have been meticulously fulfilled. The resulting infrastructure provides a secure, high-performance foundation capable of supporting the lounge's immediate entertainment needs and future business growth.

I. SOFTWARE TOOL AND PROJECT PROFILE

I. **Cisco Packet Tracer:** The primary simulation environment used to build the physical and logical topology, configure the 802.1Q trunking protocols, and perform end-to-end connectivity testing.

II. **Repository Link:** https://github.com/MateyeSelota/CMPG325-2026-125